

MODULE 4

MARKET STRUCTURE & PRICING PRACTICES

Market is a place where the various sellers and buyers sell and buy the products respectively. In this module we will be discussing about various market structure.

Perfect Competition: Perfect market is a market condition with large buyers and sellers dealing in homogenous commodity or identical product and in this market condition, all buyers and sellers are aware of the price of the product that indicates the price of the product throughout the market.

Perfect competition describes a market structure where competition is at its greatest possible level. To make it more clear, a market which exhibits the following characteristics in its structure is said to show perfect competition:

1. Large number of buyers and sellers
2. Homogenous product is produced by every firm
3. Free entry and exit of firms

Features of Perfect Competition:

1. Large number of sellers and buyers: In a perfectly competitive market, there are large numbers of buyers and seller each demanding a small part of the total market demand and supply of the product respectively. As a result, no single seller or buyer is in a position to influence the market price determined by the forces of market demand and supply.

2. Homogeneous Product: In a perfectly competitive market, all the firms produce and supply the identical products. It means that the products of all the firms are perfect substitutes of each other. As a result of this, the price elasticity of demand for a firm's product is infinite.

3. Free entry and exit: In a perfectly competitive market, there are no restrictions on the entry of new firms into market or on the exit of existing firms from the market.

4. Perfect knowledge: In a perfectly competitive market, the firms and the buyers possess perfect information about the market. It implies that no buyer or firm is ignorant about the price prevailing in the market.

5. Perfect mobility of factors of production: In a perfectly competitive market, the factors of production are completely mobile leading to factor-price equalization throughout the market

6. Free and Perfect Competition among buyers and sellers: In a perfect market, there are no checks either on the buyers or sellers. They are free to buy or to sell to any person. It means there are no monopolies.

7. Absence of Price control: In a perfectly competitive market, each buyer is aware of product and its price in the market likewise the seller is also aware of the pricing of the product in the market. Thus no individual in the market can influence the market.

8. Absence of transport cost: In a perfectly competitive market, it is assumed that there are no transport costs. If transport costs are incurred, prices should be different in different sectors of the market.

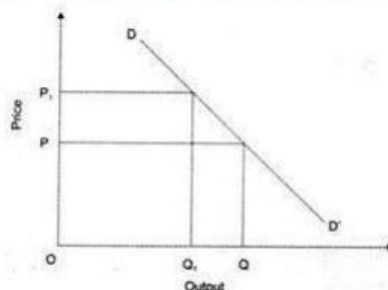
9. One price of Commodity: As there are large sellers in the market, all the sellers sell the product at same price. Every firm is a price taker. It takes the price as decided by the forces of demand and supply. No firm can influence the price of the product.

10. Independent Relationship among buyers and sellers: In a perfectly competitive market, it is considered to be no seller is related to buyer. Thus there is no chance for buyer to ask the product for lesser price and buyer will not sell the products at lesser price.

Determination of Price in Perfect Competition:

In perfect competition, the price of a product is determined at a point at which the demand and supply curve intersect each other. This point is known as **equilibrium point** as well as the price is known as **equilibrium price**. In addition, at this point, the quantity demanded and supplied is called **equilibrium quantity**.

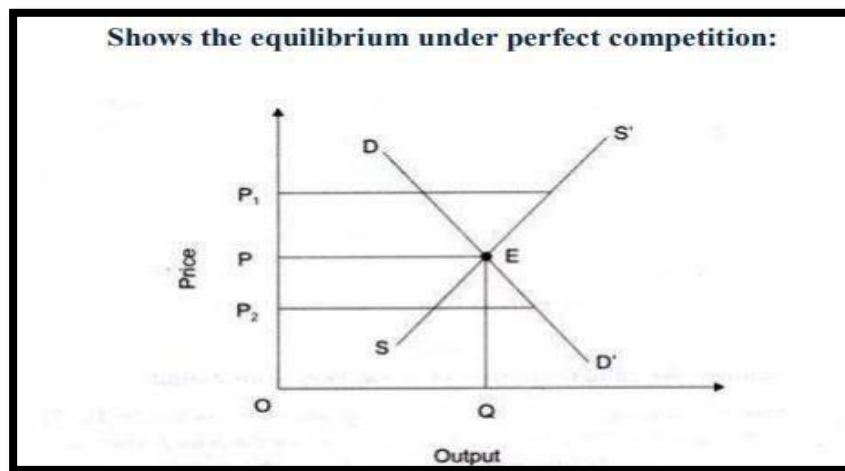
Represents the demand curve under perfect competition:



From the above figure when price is OP , the quantity demanded is OQ . On the other hand, when price increases to OP_1 , the quantity demanded reduces to OQ_1 . Therefore, under perfect competition, the demands curve (DD') slopes downward

From the above figure the quantity supplied is OQ at price OP . When price increases to OP_1 , the quantity supplied increases to OQ_1 . This is because the producers are able to earn large profits by supplying products at higher price. Therefore, under perfect competition, the supply curves (SS') slopes upward.

From the Below figure, it can be seen that at price OP_1 , supply is more than the demand.



Therefore, prices will fall down to OP . Similarly, at price OP_2 , demand is more than the supply. Similarly, in such a case, the prices will rise to OP . Thus, E is the equilibrium at which equilibrium price is OP and equilibrium quantity is OQ .

The price at which demand and supply are equal is known as equilibrium price and the quantity bought and sold at the equilibrium price is known as equilibrium output.

In the figure, equilibrium price is determined at the point E where both demand and supply are equal. The upper limit of the price of a product is determined by the demand. The lower limit of the price is determined by the production cost. The point E can be regarded as the position of stable equilibrium.

1. Very Short Period
2. Short Period
3. Long Period

1. Very Short Period:

Refers to a time period in which quantity supplied of a product cannot be increased with increase in its demand.

In simple terms, in very short period of time, the supply of a product is fixed. For example, a confectioner has 20 pastries at a particular time.

After an hour, a customer requires 40 pieces of pastries. In such a case, the confectioner cannot prepare 20 more pastries in an hour and can only supply 20 pastries. Therefore, the supply is fixed, which is 20 in the given example. The price determined in very short period is known as market price.

2. Short Period:

Refers to a time period in which the level of supply of a particular product can be increased, but only as per the production capacity of an organization. For example, an organization can produce 50 mobile phones in a day. This is the maximum production capacity of the organization.

Suppose the demand of mobile phones increases to 150 per day for three days. In such a scenario, the organization cannot install new machines or hire more labor in three days to meet the additional demand. Thus, the supply is fixed even in short period and the price determined in this period is known as sub-normal price.

3. Long Period:

Refers to a period in which the supply of a product can be increased or decreased with the changing level of demand. In this period, organizations can install new machines or hire more labor to meet the supply requirements.

Generally, in the long period, demand is determined by change in the size of population and change in customer's tastes and preferences. On the other hand, organizations can reduce production level if there is decrease in demand. Therefore, it can be said that in the long period, the price of a product is influenced by supply. The price in the long period is called normal price.

MONOPOLY:

A market structure characterized by a single seller, selling a unique product in the market. In a monopoly market, the seller faces no competition, as he is the sole seller of goods with no close substitute.

"Monopoly is a market situation in which there is a single seller. There are no close substitutes of the commodity it produces, there are barriers to entry". -Koutsoyiannis

Features:

- 1. One Seller and Large Number of Buyers:** The monopolist's firm is the only firm; it is an industry. But the number of buyers is assumed to be large.
- 2. No Close Substitutes:** There shall not be any close substitutes for the product sold by the monopolist. The cross elasticity of demand between the product of the monopolist and others must be negligible or zero.
- 3. Difficulty of Entry of New Firms:** There are either natural or artificial restrictions on the entry of firms into the industry, even when the firm is making abnormal profits.
- 4. Monopoly is also an Industry:** Under monopoly there is only one firm which constitutes the industry. Difference between firm and industry comes to an end.
- 5. Price Maker:** Under monopoly, monopolist has full control over the supply of the commodity. But due to large number of buyers, demand of any one buyer constitutes an infinitely small part of the total demand. Therefore, buyers have to pay the price fixed by the monopolist.
- 6. Price discrimination:** A company that is operating in a monopolistic market can change the price and quantity of the product or service. Price discrimination occurs when the company sells the same product to different buyers at different prices.

Considering that the market is elastic, the company will sell a higher quantity of the product if the price is low and will sell a lesser quantity if the price is high.

7. No Competition: As there is only one firm in monopoly there are no competition in this type of markets. Thus, this market is competition free.

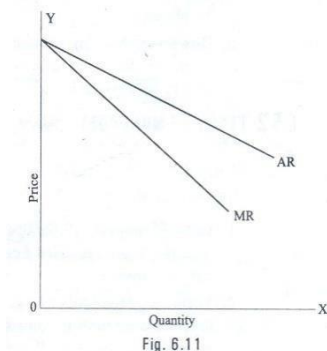
8. Full Control over Price: As a single firm itself acts as the industry, the firm operating in this market will always have an hedge of pricing the products at its will. Thus it controls the price completely.

9. Full Control over Supply: In this market the products or services do not have close substitutes and thus the producers have the control over the supply of their products.

Pricing or Price-Output Determination Under Monopoly:

Pricing under Monopoly

Monopoly refers to a market situation where there is only one seller. He has complete control over the supply of a commodity. He is therefore in a position to fix any price. Under monopoly there is no distinction between a firm and an industry. This is because the entire industry consists of a single firm.



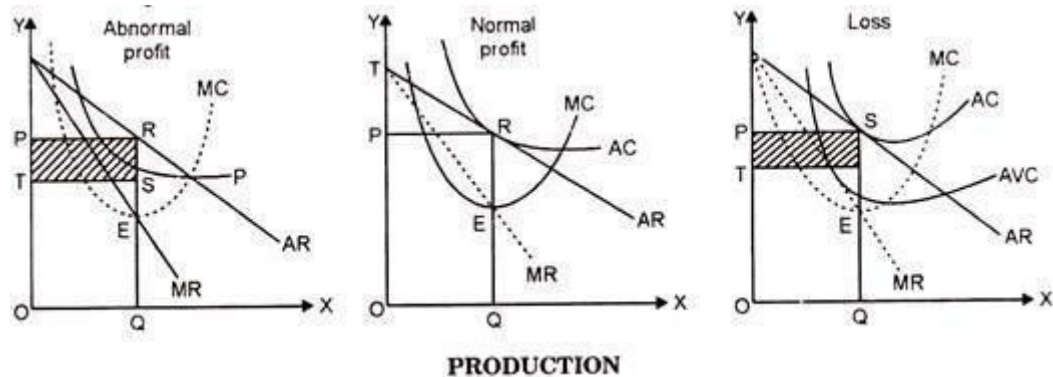
Determination of Price in the SHORT Period:

If, in a short period, the cost of production of a commodity is zero, he will go on producing it to the extent or so long the marginal revenue from the sale of that commodity does not fall to zero. As soon as the marginal revenue is zero he will not increase its supply.

Some economists think that, in a short period, three different situations may arise before the monopolist:

- (i) When the monopolist earns abnormal profits,
- (ii) When he gets only normal profits, and
- (iii) When he suffers losses.

The explanation and diagrams of these situations are given below:



On the point E the firm is in equilibrium when $MC = MR$. Thereafter MC curve starts to rise. Under the condition, OP is the price and OQ is the 'total production' of the commodity so determined. In order to calculate profits or losses, we will have to measure the difference between AR and AC. If $AR > AC$, the difference between the two is profit per unit and by multiply it with total number of units produced we can get total profit.

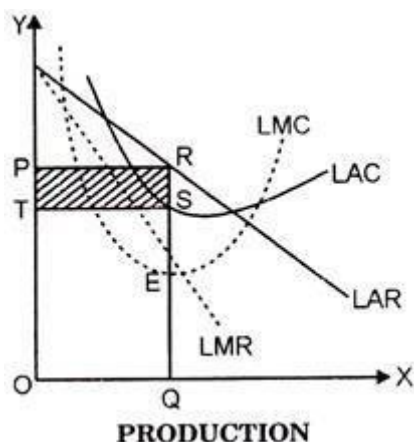
In the first figure $RQ = OP$ is the price, TO is the cost of production per unit. Thus, $RS = PT$ is unit for profit. On the OQ quantity of production, total profit is PTSR shaded area which is abnormal profit. In the second figure $RQ = OP$ is the determined price and RQ is the average cost. Under this condition, there will be only normal profit.

In the figure three also price per unit is $RQ = OP$ but cost per unit is SQ. Thus, SR (TP) is loss per unit. As a result TPRS shaded area will be the total loss. But this loss is only short period phenomenon. In the long period, this loss will disappear, under that condition and situation, only profit will be earned.

Determination of Price in the Long Period:

In the long period the monopolist introduces changes in his equipment's and techniques of production. During this period in order to gain excess profit, he will change efficiency and capacity of his resources according to his need. But the determination of the quantity of production follows, the same line as under short period.

This is clear from the following figure:



In this figure LMC and LMR intersect each other at the point E and after that LMC goes on

rising. Thus OQ production is determined and OP is the price. But average cost is SQ. So profit per unit is RS and at OQ output the total profit is PTRS.

Under Price Competition $AR = MR$, whereas under Monopoly $MR < AR$.

Under perfect competition price is determined by the interaction of total demand and supply. This price is acceptable to all the firms in the industry. No firm can change this price. So, average revenue and marginal revenue, at every level of production, will be constant and equal. Their curves are parallel to X-axis.

Under Monopoly, to sell every additional unit of the commodity price will have to be lower. In this way, with the sale of every additional unit, average and marginal income goes on falling. But the decrease in average revenue is relatively less sharp than the decrease in marginal revenue. It is because marginal revenue is limited to one unit, whereas in case of average revenue, the decrease price is divided by the number of units. Therefore, the fall in average revenue has relatively less slope. That is the reason why marginal revenue is less than average revenue.

PRICE DISCRIMINATION:

In monopoly, there is a single seller of a product called monopolist. The monopolist has control over pricing, demand, and supply decisions, thus, sets prices in a way, so that maximum profit can be earned. The monopolist often charges different prices from different consumers for the same product. This practice of charging different prices for identical product is called price discrimination.

According to Robinson, "Price discrimination is charging different prices for the same product or same price for the differentiated product."

Different price discrimination are as follows:

First Degree Price Discrimination
Second Degree Price Discrimination
Third Degree Price Discrimination

- i First-degree Price Discrimination:** Refers to a price discrimination in which a monopolist charges the maximum price that each buyer is willing to pay. This is also known as perfect price
- ii discrimination** as it involves maximum exploitation of consumers. In this, consumers fail to enjoy any consumer surplus. First degree is practiced by lawyers and doctors.

iii. Second-degree Price Discrimination: Refers to a price discrimination in which buyers are divided into different groups and different prices are charged from these groups depending upon what they are willing to pay. Railways and airlines practice this type of price discrimination.

iv. Third-degree Price Discrimination: Refers to a price discrimination in which the monopolist divides the entire market into submarkets and different prices are charged in each submarket. Therefore, third-degree price discrimination is also termed as market segmentation.

In this type of price discrimination, the monopolist is required to segment market in a manner, so that products sold in one market cannot be resold in another market. Moreover, he/she should identify the price elasticity of demand of different submarkets. The groups are divided according to age, sex, and location. For instance, railways charge lower fares from senior citizens. Students get discount in cinemas, museums, and historical monuments.

MONOPOLISTIC COMPETITION

Monopolistic competition is a form of market structure having large number of sellers and buyers dealing in similar products or close substitute products where there are no entry or exit barriers.

Features of Monopolistic Competition:

1. Large Number of Sellers:

There are large numbers of firms selling closely related, but not homogeneous products. Each firm acts independently and has a limited share of the market. So, an individual firm has limited control over the market price. Large number of firms leads to competition in the market.

2. Product Differentiation:

Each firm is in a position to exercise some degree of monopoly (in spite of large number of sellers) through product differentiation. Product differentiation refers to differentiating the products on the basis of brand, size, colour, shape, etc. The product of a firm is close, but not perfect substitute of other firm.

Implication of 'Product differentiation' is that buyers of a product differentiate between the same products produced by different firms. Therefore, they are also willing to pay different prices for the same product produced by different firms. This gives some monopoly power to an individual firm to influence market price of its product.

3. Selling costs on Advertisements:

Under monopolistic competition, products are differentiated and these differences are made known to the buyers through selling costs. Selling costs refer to the expenses incurred on marketing, sales promotion and advertisement of the product. Such costs are incurred to persuade the buyers to buy a particular brand of the product in preference to competitor's brand. Due to this reason, selling costs constitute a substantial part of the total cost under monopolistic competition.

4. Freedom of Entry and Exit:

Under monopolistic competition, firms are free to enter into or exit from the industry at any time they wish. It ensures that there are neither abnormal profits nor any abnormal losses to a firm in the long run. However, it must be noted that entry under monopolistic competition is not as easy and free as under perfect competition.

5. Lack of Perfect Knowledge:

Buyers and sellers do not have perfect knowledge about the market conditions. Selling costs create artificial superiority in the minds of the consumers and it becomes very difficult for a consumer to evaluate different products available in the market. As a result, a particular product (although highly priced) is preferred by the consumers even if other less priced products are of same quality.

6. Pricing Maker:

A firm under monopolistic competition is neither a price-taker nor a price-maker. However, by producing a unique product or establishing a particular reputation, each firm has partial control over the price. The extent of power to control price depends upon how strongly the buyers are attached to his brand.

7. Non-Price Competition:

In addition to price competition, non-price competition also exists under monopolistic competition. Non-Price Competition refers to competing with other firms by offering free gifts, making favourable credit terms, etc., without changing prices of their own products.

Firms under monopolistic competition compete in a number of ways to attract customers. They use both Price Competition (competing with other firms by reducing price of the product) and Non-Price Competition to promote their sales.

8. Demand curve is downward sloping:

Under monopolistic competition, large number of firms selling closely related but differentiated products makes the demand curve downward sloping. It implies that a firm can sell more output only by reducing the price of its product.

9. No Close Substitute products:

Products in this market do not have close substitutes to the competitors' products.

Price-Determination under Monopolistic Market in Short Run:

During short period there may be three situations of the firms under monopolistic competition as given below:

1. Profit Making Situation: $AR > AC$

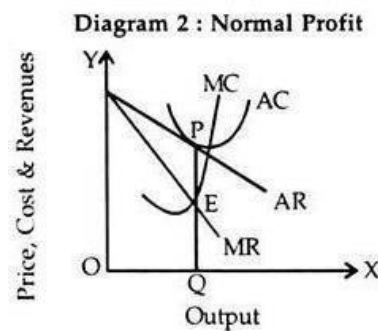
Profit making situation will be when individual firm's revenue is greater than its cost ($AR > AC$).

Profit making situation is also called abnormal or super profit situation as given in Diagram 1.

Price, costs and revenues are shown on OY-axis while output on OX-axis. The point of equilibrium is E where marginal cost is equal to marginal revenue ($MC=MR$) of the firm. The price is OP, output is OQ. The average profit ($AR-AC$) is TS and total profit is PLST.

2. Normal Profit: $AR=AC$

When the firm's average revenue is equal to its average cost the situation is called normal profit.



In the diagram output is shown on OX-axis, price, costs and revenue are shown on OY-axis. The point of equilibrium is E where firm's marginal cost is equal to its marginal revenue ($MC=MR$). Price is PQ and output is OQ. At P the average revenue is equal to average cost hence the firm is earning normal profit only.

3. Loss Making Situation: $AR < AC$

Under monopolistic situation during short period a firm will earn loss when its cost is greater than its revenue ($AC > AR$). It can be explained with the help of the Diagram 3.

Price, Cost & Revenues

Y

L

P

O

Q

Output

X

MC

AC

AR

MR

S

E

T

Price = OP

Output = OQ

Average Loss (AC-AR) = ST

Total Loss = LPTS

Price-Determination under Monopolistic Market in long Run:

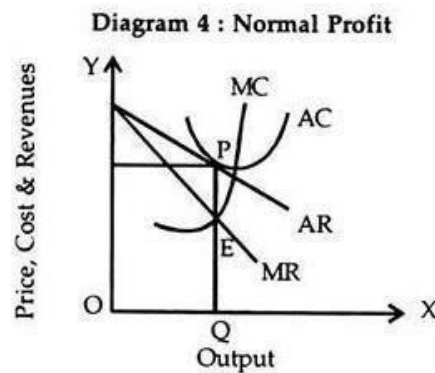
Conditions for the Equilibrium of an individual firm

1. $MC = MR$
2. The MC curve cuts the MR curve from below.

Under monopolistic competition, firms have freedom to enter and exit the industry. In the long run if firms are earning profit new firms are attracted and it will increase the output and consequently prices will fall leading to conversion of profit making situation into normal profit situation.

Contrary to it when firms are incurring losses during long period they will leave the industry. It will reduce the volume of output, prices will increase and the loss making situation will be converted into normal profit. Thus, the firms will earn normal profit only during long period. It can be seen from Diagram 4.

Price, costs and revenue are shown on OY-axis and output on OX-axis. Point of equilibrium ($MC=MR$) is E. Price is PQ and output is OQ. At P point average cost is equal to average revenue ($AC=AR$). Hence, the firm is earning normal profit only during long period.



PRODUCT DIFFERENTIATION:

A firm may have convinced consumers that its product is significantly better than the product of new entrants. The new firm may be forced to sell at lower price and reduce profit though the existing product may not essentially be superior.

The real differentiation refers to the technical features like the product's technical life and performance, durability, cost of operation and maintenance, etc.

On the other hand, the non-technical differentiation may take the form of brand names, trademark, packing, shape, size, etc. The non-technical differentiation adds a subjective appeal to a product inducing buyers to increase its demand or pay more for it.

Price: Price can be used to differentiate a product in two ways. Companies can charge the lowest price compared to competitors to attract cost-conscious buyers—the retailer Costco is an example. However, companies can also charge high prices to imply quality and that a product is a luxury or high-end item, such as a Bugatti sports car.

Performance and Reliability: Products can be differentiated based on their reliability and durability. Some batteries, for example, are reputed to have a longer life than other batteries, and consumers will buy them based on this factor.

Location and Service: Local businesses can differentiate themselves from their larger national competitors by emphasizing that they support the local community. A local restaurant, for example, will hire locally and may source its food and ingredients from local farmers and purveyors.

Quality: How does the quality, reliability, and ruggedness of your product compare to others on the market?

Design: Have you done something different with your design? Is it minimalistic and sleek? Easy-to-navigate?

Customization: Can you customize parts of the product that competitors cannot?

Similarly we can differentiate the products on basis of color, brand, size etc.

OLIGOPOLY:

Oligopoly market is a situation where in there are few number of sellers and large buyers dealing in homogenous product or differentiated products and there are entry-exit barriers for new firm.

Features of Oligopoly:

1 Few firms: Under oligopoly, there are few large firms. The exact number of firms is not defined. Each firm produces a significant portion of the total output. There exists severe competition among different firms and each firm try to manipulate both prices and volume of production to outsmart each other. For example, the market for automobiles in India is an oligopolist structure as there are only few producers of automobiles.

2 Interdependence: Firms under oligopoly are interdependent. Interdependence means that actions of one firm affect the actions of other firms. A firm considers the action and reaction of the rival firms while determining its price and output levels. A change in output or price by one firm evokes reaction from other firms operating in the market.

3 Nature of the Product: The firms under oligopoly may produce homogeneous or differentiated product.

If the firms produce a homogeneous product, like cement or steel, the industry is called a pure or perfect oligopoly.

If the firms produce a differentiated product, like automobiles, the industry is called differentiated or imperfect oligopoly.

4 Barriers to Entry of Firms: The main reason for few firms under oligopoly is the barriers, which prevent entry of new firms into the industry. Patents, requirement of large capital, control over crucial raw materials, etc, are some of the reasons, which prevent new firms from entering into industry. Only those firms enter into the industry, which is able to cross these barriers. As a result, firms can earn abnormal profits in the long run.

5 Non-Price Competition: Under oligopoly, firms are in a position to influence the prices. However, they try to avoid price competition for the fear of price war. They follow the policy of price rigidity. Price rigidity refers to a situation in which price tends to stay fixed irrespective of changes in demand and supply conditions. Firms use other methods like advertising, better services to customers, etc. to compete with each other.

If a firm tries to reduce the price, the rivals will also react by reducing their prices. However, if it tries to raise the price, other firms might not do so. It will lead to loss of customers for the firm, which intended to raise the price. So, firms prefer non- price competition instead of price competition.

6 Lack of Uniformity: Another feature of oligopoly market is the lack of uniformity in the size of firms. Firms differ considerably in size. Some may be small, others very large. Such a situation is asymmetrical. This is very common in the American economy. A symmetrical situation with firms of a uniform size is rare.

7. Existence of Price Rigidity: In oligopoly situation, each firm has to stick to its price. If any firm tries to reduce its price, the rival firms will retaliate by a higher reduction in their prices. This will lead to a situation of price war which benefits none. On the other hand, if any firm increases its price with a view to increase its profits; the other rival firms will not follow the same. Hence, no firm would like to reduce the price or to increase the price. The price rigidity will take place.

8 Advertising: As the firms follow non-price competition by keeping the prices close to rival firms; to increase the sales and profit of the firm they have to involve themselves in marketing and promotions of the product.

9 Selling Costs: Since firms try to avoid price competition and there is a huge interdependence among firms, selling costs are highly important for competing against rival firms for a larger market share.

Price Determination under Oligopoly:

Oligopoly can be either collusive or non-collusive. Collusive oligopoly is a market situation wherein the firms cooperate with each other in determining price or output or both. A non-collusive oligopoly refers to a market situation where the firms compete with each other rather than cooperating.

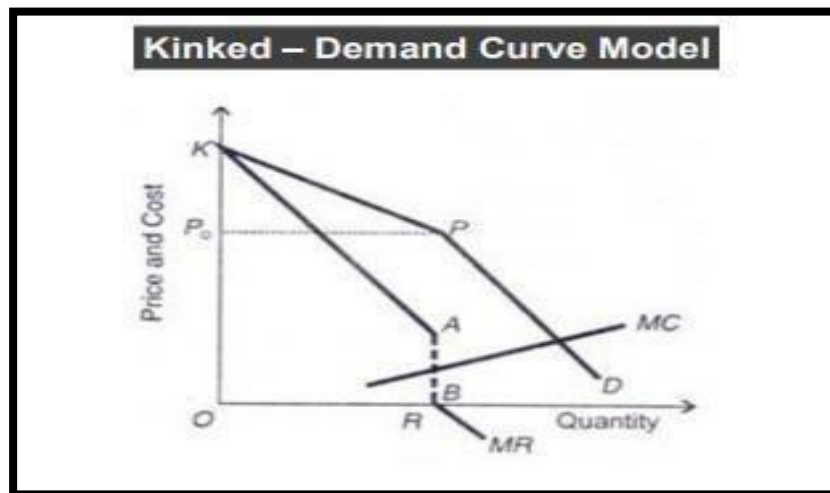
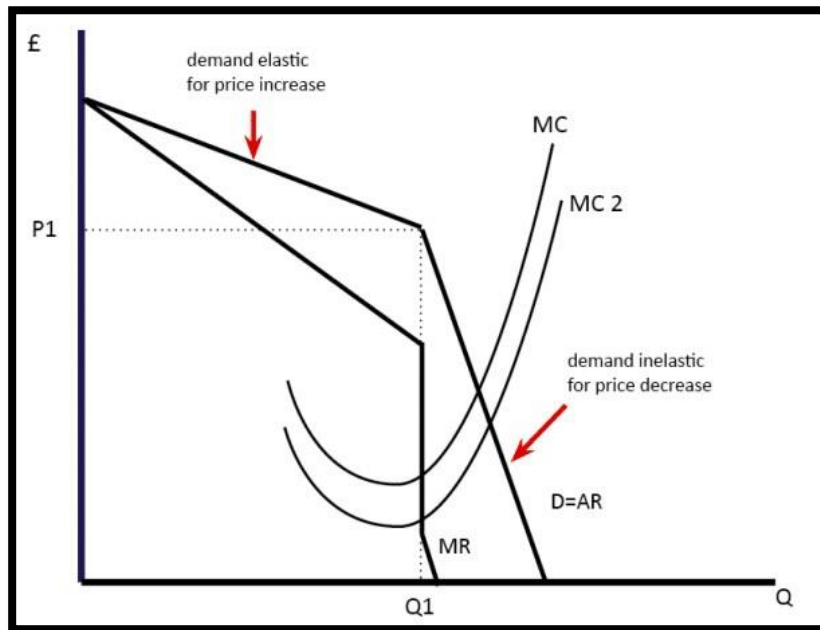
1. Price Determination in Non-Collusive Oligopoly:

In this case, each firm follows an independent price and output policy on the basis of its judgment about the reactions of his rivals. If the firms are producing homogeneous products, price war may occur. Each firm has to fix the price at the competitive level. On the contrary, in case of differentiated oligopoly, due to product differentiation, each firm has some monopoly control over the market and therefore charge near monopoly price.

Usually, in Oligopolistic markets, there are many price rigidities. In 1939, Paul Sweezy used an unconventional demand curve – the kinked demand curve to explain these rigidities.

Kinked-Demand Curve:

Like traditional demand curves, kinked demand curves are downward sloping. As the name suggests, the kinked demand curves have a 'kink'. This kink is nothing but a discontinuity at a concave bend and this kink is what sets it apart from the traditional demand curves. Now let's find out why these kinks exist in the first place.



MR = Marginal Revenue, MC = Marginal Cost

KPD = Deman Curve or Average Revenue, OR is output at price OP_0

Assumptions:

1. Firm selling their products at satisfactory price.
2. Rivals will tend not to raise prices in response to the firm's higher prices.
3. Rivals will not follow an attempted increase in price by one of the firms but will follow a decrease.

In the figure above, KPD is the is the kinked-demand curve and OP_0 is the prevailing price in the oligopoly market for the OR product of one seller. Starting from point P, corresponding to the point OP_1 , any increase in price above it will considerably reduce his sales as his rivals will not follow his price increase.

This is because the KP portion of the curve is elastic and the corresponding portion of the MR curve (KA) is positive. Therefore, any price increase will not just reduce the total sales but also his total revenue and profit. On the other hand, if the seller reduces the price of the product below OPQ (or P), his rivals will also reduce their prices.

However, even if his sales increase, his profits would be less than before. This is because the PD portion of the curve below P is less elastic and the corresponding part of the marginal revenue curve below R is negative. Therefore, in both price-raising and price-reducing situations, the seller is the loser. He will stick to the prevailing market price OP_0 which remains rigid.

This asymmetrical behavioral pattern results in a kink in the demand curve and hence there is price rigidity in oligopoly markets. The prices remain rigid at the kink (point P). In other words, the price will remain sticky at OP_0 and the output = OR at this price.

Due to the difference in the elasticities, the MR curve becomes discontinuous corresponding to the point of change in elasticity of the demand curve. The kink represents this. At the output $< OR$, the demand curve is KP and the corresponding MR curve is KA and $MR > MC$. For output $> OR$, the demand curve is PD and the corresponding MR curve is BMR and $MR < MC$.

2. Price Determination in Collusive Oligopoly:

Sometimes, firms may try to remove uncertainty related to acting independently and enter into price agreements with each other. This is collusion. Collusion is either formal or informal. It can take the form of cartel or price leadership.

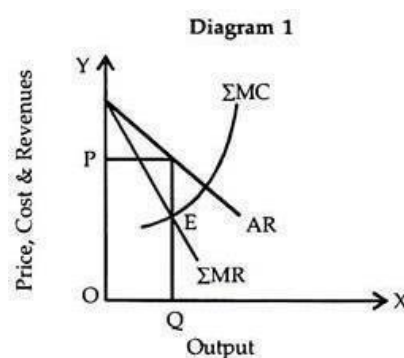
Price and Output under Perfect Collusion:

Under oligopoly, perfect collusion may be formed among different producers, sellers in two ways, namely, centralised cartel and market sharing cartel, and price and output are determined accordingly.

A cartel is an association of independent firms within the same industry which follow the common policies relating to price, output, sale, profit maximization, and the distribution of products.

(a) Centralised Cartel:

Under this type of collusion different firms of oligopoly market structure set up a centralised cartel. These firms transfer their function relating to managerial decisions and other activities to the centralised cartel to improve the volume of profit. Centralised cartel aims at maximisation of profit of all the firms. The cartel fixes price of the product, volume of output and production quota of individual firms.



In the diagram AR is the demand curve of industry and MR is the marginal revenue curve which has been drawn on the basis of AR or demand curve. ΣMC is the total marginal cost curve of the

industry which is the aggregate marginal cost of two firms engaged in production in that industry. Marginal revenue curve of industry is cut by the marginal cost curve of the industry at point E where $\sum MR$ is equal to $\sum MC$ ($\sum MR = \sum MC$). It is the equilibrium of the industry. Price is OP and output is OQ in the industry. Number of products to be produced by a firm is fixed on the basis of MR, that is when MR of firm is equal to combined MR of market.

(b) Market Sharing Cartel:

Another form of perfect collusion is the market sharing by the firms. This type of collusion can be effective and successful when all the firms are producing homogeneous product and production costs are similar. In order to explain this type of collusion we assume that there are two firms producing on the uniform cost of production and are ready to share market on 50:50 basis.

There are two main methods of market-sharing:

(a) Non-price competition; and (b) Quota system.

(a) Non-Price Competition Cartel:

The non-price competition agreement among oligopolistic firms is a loose form of cartel. Under this type of cartel, the low-cost firms press for a low price and the high-cost firms for a high price. But ultimately, they agree upon a common price below which they will not sell. Such a price must allow them some profits. The firms can compete with one another on a non-price basis by varying the colour, design, shape packing etc. of their product and having their own different advertising and other selling activities. Thus each firm shares the market on a non-prices basis while selling the product at the agreed common price.

(b) Market Sharing by Quota Agreement:

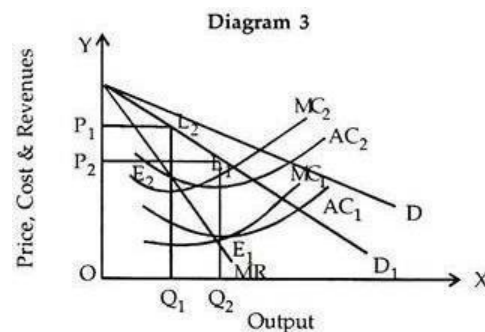
The second method of market sharing is the quota agreement among firms. (All firms in an oligopolistic industry enter into collusion for charging an agreed uniform price. But the main agreement relates to the sharing of the market equally among member firms so that each firm gets profits on its sales.

Price and Output under Imperfect Collusion: Price Leadership

Under oligopoly price and output can also be determined without any collusion among the firms. The firms may decide to follow a firm in price and output determination in the long run. Such sort of policy is called price leadership under oligopoly.

Price leadership is based on informed collusion. Under price leadership, one firm is a large or dominant firm and acts as the price leader who fixes the price for the products while the other firms allow it.

In the diagram output is shown on OX-axis while price, cost, and revenue are shown on OY-axis. DD is the market demand curve while DD₁ is the demand curve of the firm. AC₂ and MC₂ are average cost curve and marginal cost curve of the firm having high cost of production. Its point of equilibrium is E₂ where the price is OP₂ and the output is OQ₂.



The average cost curve (AC₁) and marginal cost curve (MC₁) are of a firm having least cost of production. Its point of equilibrium is E₁ where the price of the firm is OP₁ and output is OQ₁. The marginal revenue curve (MR) of both the firms is equal to their MC₁ and MC₂. The firm producing OQ₁ output with OP₁ price is the low cost producing firm and it will be the price leader in the market and the same price policy OP₁ will be following by other firm.

There are several types of price leadership. The following are the principal types:

- (a) **Price leadership of a dominant firm**, i.e., the firm which produces the bulk of the product of the industry. It sets the price and rest of the firms simply accepts this price.
- (b) **Barometric price leadership**, i.e., the price leadership of an old, experienced and the largest firm assumes the role of a leader, but undertakes also to protect the interest of all firms instead of promoting its own interests as in the case of price leadership of a dominant firm.

(c) **Exploitative or Aggressive price leadership**, i.e., one big firm built its supremacy in the market by following aggressive price leadership. It compels other firms to follow it and accept the price fixed by it. In case the other firms show any independence, this firm threatens them and coerces them to follow its leadership.

Descriptive Pricing Approaches:

Pricing is the art of converting the value of products or services into quantitative terms to customers at a point of time.

Factors determining Pricing:

(i) **Cost of Production:** Cost of production is the main component of price. No company can sell its product or services at less than the cost of production. Thus, before price fixation, it is necessary to compile data relating to the cost of production and keep that in mind.

(ii) **Demand for Product:** Intensive study of demand for product and services in the market be undertaken before price fixation. If demand is relatively more than supply, higher price can be fixed.

(iii) **Price of Competing Firms:** It is necessary to take into consideration prices of the products of the competing firms prior to fixing the price. In case of cut-throat competition it is desirable to keep prices low.

(iv) **Purchasing Power of Customers:** What are the purchasing power of the customers and at what price and how much they can purchase? It should also be taken into consideration.

(v) **Government Regulation:** If the price of the commodity and services is to be fixed as per the regulation of the government, it should also be borne in mind.

(vi) **Objective:** Usually, at the time of price fixation a certain amount of profit is added to the cost of production. If company's objective is to earn higher profit it may add higher amount of it.

(vii) **Marketing Method Used:** Price is also influenced by the marketing method used by the company, e.g., commission which is to be paid to the middlemen for sale of the goods is also added to the price. Similarly, if the customers are to be provided "after sale service" facility, then those expenses are also added to the price.

PRICING STRATEGIES ARE:

- Full cost pricing
- Product line pricing
- Price Skimming
- Penetration Pricing
- Loss leader pricing
- Peak Load pricing

1. Full cost pricing: Full cost pricing is a practice where the price of a product is calculated by a firm based on its direct costs per unit of output plus a markup to cover overhead costs and profits.

It is a price-setting method under which you add together the direct material cost, direct labor cost, selling and administrative costs, and overhead costs for a product, and add to it a markup percentage (to create a profit margin) in order to derive the price of the product. The pricing formula is:

(Total production costs + Selling and administration costs +Markup)

Number of units expected to sell

2. Product line pricing: Product line pricing involves the separation of goods and services into cost categories in order to create various perceived quality levels in the minds of consumers. You might also hear product line pricing referred to as price lining, but they refer to the same practice.

3. Price Skimming: Price skimming is a product pricing strategy by which a firm charges the highest initial price that customers will pay and then lowers it over time. As the demand of the first customers is satisfied and competition enters the market, the firm lowers the price to attract another, more price sensitive segment of the population. The skimming strategy gets its name from "skimming" successive layers of cream, or customer segments, as prices are lowered over time.

4. Penetration Pricing: Penetration pricing is a marketing strategy used by businesses to attract customers to a new product or service by offering a lower price during its initial offering. The

lower price helps a new product or service penetrate the market and attract customers away from competitors. Market penetration pricing relies on the strategy of using low prices initially to make a wide number of customers aware of a new product.

The goal of a price penetration strategy is to entice customers to try a new product and build market share with the hope of keeping the new customers once prices rise back to normal levels.

5. Loss leader pricing: Loss leader pricing is an aggressive pricing strategy in which a store sells selected goods below cost in order to attract customers who will, according to the loss leader philosophy, make up for the losses on highlighted products with additional purchases of profitable goods. Loss leader pricing is employed by retail businesses; a somewhat similar strategy sometimes employed by manufacturers is known as penetration pricing.

6. Peak Load pricing: The Peak Load Pricing is the pricing strategy wherein the high price is charged for the goods and services during times when their demand is at peak. In other words, the high price charged during the high demand period is called as the peak load pricing.

This type of price discrimination is based on the efficiency, i.e. a firm discriminates based on high usage, high-traffic, high demand times and low demand times. The consumer who purchases the commodity during the high demand period has to pay more as compared to the one who buys during low demand periods.

Question Bank:

1. Define perfect competition.
2. Define cartels.
3. What is monopoly? Discuss its characteristics.
4. What is price discrimination? Mention the degrees of price discrimination.
5. Mention the characteristics of perfect competition.
6. Explain the concept of monopolistic competition. How prices are determined.
7. Elaborate the characteristics of oligopoly.
8. Define market structure. What are the different types of market structure?
9. Explain kinked demand curve along with graph and its assumptions.
10. How are prices determined under perfect competition? Explain with the help of suitable graphs.
11. Discuss the various descriptive pricing strategies.
12. Enumerate the factors governing pricing decisions.